Lightweight Coronary Artery Segmentation via Knowledge Distillation: Overcoming Resolution and Prompt Artifacts for Edge Deployment

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Abstract

Real-time coronary artery segmentation during percutaneous coronary intervention (PCI) requires models that operate under severe compute constraints without dedicated GPU infrastructure. We present CA-SAM2, a knowledge distillation pipeline compressing a fine-tuned SAM2.1 teacher (Hiera-Tiny, 38.9M params) into MobileSAM (ViT-Tiny, 10.1M params). We investigate two critical methodological pathologies in medical foundation model distillation: (1) an input resolution confound, where conventional 512×512 evaluation artificially penalizes the teacher on thin tubular structures (1–4 px wide), and (2) a prompt-synchronization artifact, where distilling with static centroid prompts while querying students with jittered clicks causes severe supervision conflict.

By enforcing native 1024×1024 resolution parity and dynamic prompt synchronization, Response Knowledge Distillation recovers from a previously reported 0.687 to Dice 0.7973 ± 0.0754 (IoU 0.6691) on 200 held-out ARCADE test images, effectively matching the 38.9M teacher (0.7995 ± 0.0751) at one-fourth the parameters and memory footprint. Crucially, on an external, out-of-distribution clinical dataset of 134 unseen coronary angiograms (DCA1, 300×300 px), the 10.1M student statistically significantly outperforms its teacher (Dice 0.6539 ± 0.0834 vs. 0.6359 ± 0.0785, $p = 0.0041$) with a 4.1% gain in precision (0.6275 vs. 0.5861), showing that compact student representations resist background contrast overfitting under domain shift. Finally, on continuous fluoroscopy video sequences (Coronary-Dominance), the student executes at 33.2 FPS (30.1 ms/frame) with robust contrast wash-in tracking, establishing feasibility for point-of-care cath-lab deployment.

1 Introduction

Coronary artery disease remains the leading cause of mortality worldwide. During percutaneous coronary intervention (PCI) in the cardiac catheterization laboratory, interventional cardiologists rely on real-time X-ray fluoroscopy to navigate guidewires, detect vessel stenosis, and position intracoronary stents. While automated vessel segmentation provides crucial quantitative visual guidance, cath-lab workstations are typically constrained by low-power legacy hardware (often ≤ 8 GB system RAM and no dedicated NVIDIA GPU), precluding the deployment of high-parameter vision foundation models.

Recently, the Segment Anything Model (SAM) [Kirillov et al., 2023] and its video successor SAM2 [Ravi et al., 2024] have demonstrated remarkable interactive zero-shot segmentation capabilities. Domain-specific medical adaptations, such as MedSAM [Ma et al., 2024] and MedSAM2 [Zhu et al., 2025], yield promising results across CT, MRI, and ultrasound. However, two fundamental obstacles hinder their edge deployment in interventional fluoroscopy:

1. Model Footprint and Latency: Standard SAM2 architectures require substantial GPU memory and compute, exceeding edge cath-lab capabilities.

2. Resolution Constraints on Thin Vessels: Coronary vessels in fluoroscopic projections are often only 1–4 pixels wide at standard 512×512 processing resolutions, causing distal vessel branches to fall below the receptive Nyquist rate in deep strided feature stages.

To bridge this gap, knowledge distillation (KD) [Hinton et al., 2015] offers a principled framework to compress heavy teachers into edge-efficient students such as MobileSAM [Zhang et al., 2023]. However, prior literature has observed conflicting results when applying response-based KD in medical imaging, often reporting that distilling teacher soft logits actively degrades student accuracy compared to pure ground-truth supervision.

In this work, we systematically investigate the mechanics of distilling SAM2 into MobileSAM for coronary angiography. Our contributions are fourfold:

- Resolution Confound Resolution: We demonstrate that reported margins where lightweight students apparently "surpassed" their larger teachers were

driven by an input dimension asymmetry (1024×1024 student vs. 512×512 teacher). At equal 1024×1024 resolution, the teacher scores 0.7995 Dice, matching student capacity.

- Prompt Synchronization Breakthrough: We identify a silent training bug in interactive model distillation: supervising student predictions against static teacher prompt caches while querying the student with randomized clicks penalizes correct prompt-conditioned outputs. Dynamically synchronizing prompt coordinates lifts Response KD from 0.6869 to 0.7973 Dice.
- External Clinical Domain Shift (DCA1): On 134 unseen coronary angiograms from an independent hospital and imaging device, the 10.1M student outperforms the 38.9M teacher (0.6539 vs. 0.6359, $p = 0.0041$), demonstrating that parameter compression serves as effective implicit regularization against background texture overfitting.
- Real-Time Fluoroscopic Video Tracking: We evaluate the student on continuous fluoroscopy sequences, demonstrating 33.2 FPS inference speed and faithful tracking across dynamic contrast wash-in and wash-out cycles.

2 Related Work

Coronary Vessel Segmentation. Classical vessel delineation relied on handcrafted multiscale Hessian filters, such as the Frangi vesselness filter [Frangi et al., 1998]. While modern deep convolutional and transformer networks have improved segmentation quality [Zhang et al., 2025, Chang et al., 2024], they typically rely on extensive pixel-level annotations. The ARCADE MICCAI benchmark [Popielarski et al., 2024] and the DCA1 database [Cervantes-Sanchez et al., 2019] represent the primary open benchmarks for coronary fluoroscopy.

Interactive Foundation Models in Healthcare. SAM [Kirillov et al., 2023] introduced promptable segmentation conditioned on point clicks and bounding boxes. MedSAM [Ma et al., 2024] fine-tuned SAM on large-scale biomedical corpora. SAM2 [Ravi et al., 2024] introduced a streaming memory module for video tracking, with MedSAM2 [Zhu et al., 2025] adapting it to volumetric medical data. SurgiSAM2 [Yue et al., 2025] and FS-MedSAM2 [Zhao et al., 2024] adapted SAM2 for surgical endoscopy. However, none addressed the architectural distillation of SAM2 into edge-deployable SAM1 models for coronary fluoroscopy.

Knowledge Distillation and MobileSAM. Response distillation [Hinton et al., 2015] transfers output probability distributions, while feature matching [Romero et al., 2014] aligns intermediate latent spaces. MobileSAM [Zhang et al., 2023] compressed SAM's heavy ViT encoder into a 10.1M TinyViT backbone using SA-1B distillation. In this work, we distill cross-architecture: transferring representations from a SAM2 Hiera-Tiny teacher into a MobileSAM TinyViT student.

3 Methodology

3.1 Architectural Formulation

Teacher (SAM2.1 Hiera-Tiny): The teacher employs a hierarchical vision transformer (Hiera-Tiny, 38.9M params) with multiscale feature pyramids ($\times 4, \times 8, \times 16, \times 32$). At native 1024×1024 input resolution, the final backbone stage produces an image embedding $\mathbf{F}_t \in \mathbb{R}^{B \times 256 \times 64 \times 64}$.

Student (MobileSAM TinyViT): The student employs a lightweight TinyViT backbone (10.1M params). Operating at native 1024×1024 resolution, TinyViT produces feature embeddings $\mathbf{F}_s \in \mathbb{R}^{B \times 256 \times 64 \times 64}$, achieving exact spatial dimension parity with the teacher without requiring lossy interpolation.

Hybrid Decoder Transplant: Because SAM1 and SAM2 decoders have architectural divergence (SAM2 introduces object-score tokens and video-memory convolutional blocks), we perform a partial structural weight transplant via non-strict state dict transfer, initializing overlapping cross-attention layers from the fine-tuned teacher while retaining standard SAM1 MLP projection blocks.

3.2 Prompt-Synchronized Distillation

Previous distillation pipelines generated static soft logits offline using a single centroid click $c_{\text{static}} = (\bar{x}, \bar{y})$ on un-augmented images. During student training, data augmentation and random prompt jittering sampled an altered coordinate $c_{\text{student}} = (\tilde{x}, \tilde{y})$. Because the mask decoder is conditioned on prompt location, minimizing:

$$\mathcal{L}_{\text{mismatch}} = \text{BCE}\left(\sigma(\hat{y}(c_{\text{student}})), \sigma(y_t(c_{\text{static}}))\right) \quad (1)$$

penalized the student for correctly segmenting vessels around its own assigned prompt, causing catastrophic optimization conflict.

To resolve this, we formulate Dynamic Prompt Synchronization: during every training iteration, the teacher is evaluated on-the-fly under torch.no_grad() on the identical augmented image and exact same prompt tuple (x, y):

$$\hat{y}_t, \mathbf{F}_t = \mathcal{M}_{\text{teacher}}(I_{\text{aug}}, (x, y)) \quad (2)$$

$$\hat{y}_s, \mathbf{F}_s = \mathcal{M}_{\text{student}}(I_{\text{aug}}, (x, y)) \quad (3)$$

3.3 Multi-Objective Loss Formulation

The overall objective optimizes ground-truth fidelity, topological centerline preservation, and teacher response matching:

$$\mathcal{L}_{\text{total}} = 0.5\mathcal{L}_{\text{KD}} + 0.5\mathcal{L}_{\text{hard}} + \lambda_{\text{feat}}\mathcal{L}_{\text{feat}} \tag{4}$$

where:

$$\begin{aligned}\mathcal{L}_{\text{hard}} = 0.5\mathcal{L}_{\text{Dice}}(\hat{y}_s, y_{\text{gt}}) + 0.2\mathcal{L}_{\text{wBCE}}(\hat{y}_s, y_{\text{gt}}) \\ + w_{\text{cl}}\mathcal{L}_{\text{clDice}}(\hat{y}_s, y_{\text{gt}})\end{aligned} \tag{5}$$

$\mathcal{L}_{\text{KD}}$ is the binary cross-entropy between student logits and teacher soft probabilities $\sigma(\hat{y}_t)$, and $\mathcal{L}_{\text{clDice}}$ is the soft centerline Dice loss [Shit et al., 2021] ensuring tubular connectivity. Because spatial dimensions are matched at 64×64, feature matching is computed directly:

$$\mathcal{L}_{\text{feat}} = \frac{1}{\Omega}\sum_{u,v}(1 + 9 \cdot y_{\text{gt}}(u,v)) \, \|\phi(\mathbf{F}_s(u,v)) - \mathbf{F}_t(u,v)\|_2^2 \tag{6}$$

where ϕ is an in-path 1×1 convolutional adapter and vessel foreground pixels receive $10\times$ weight.

4 Experiments and Results

4.1 Datasets and Implementation Details

- ARCADE Dataset: 1,200 coronary angiograms (512×512 px) from the MICCAI 2023 challenge [Popielarski et al., 2024]. Split: 900 train / 100 internal val / 200 held-out test.
- DCA1 Dataset: 134 coronary angiograms (300×300 px) with expert pixel-level ground truth from Cervantes-Sanchez et al. [Cervantes-Sanchez et al., 2019]. Used exclusively for external domain shift evaluation.
- CoronaryDominance Video: 10 multi-frame fluoroscopic cine studies (20–70 frames/study) from Danilov et al. [Danilov et al., 2025].

Training was executed on an NVIDIA Grace Blackwell GB10 GPU using AdamW (lr 3×10^{-4}, cosine decay, weight decay 0.01). Evaluation uses continuous bilinear logit upsampling before thresholding.

4.2 In-Distribution Results (ARCADE)

Table 1 shows the primary benchmarking results:

1. Resolution Confound: Fine-tuning the teacher architecture at 1024×1024 improves Dice from 0.7670 to 0.7995 (+0.0325). The student's apparent edge over the 512px teacher was predominantly an input resolution artifact.
2. Recovery of Response KD: Under static prompt mismatch (v3), Response KD collapsed to 0.6869. With dynamic prompt synchronization, Response KD achieves Dice 0.7973 ± 0.0754 (+0.1104 gain), matching the 38.9M teacher within statistical noise ($\sigma \approx 0.075$) at $4\times$ smaller size.

4.3 External Domain Shift Generalization (DCA1)

To rigorously evaluate generalization without potential benchmark contamination, we tested both models directly on all 134 unseen images of the DCA1 dataset without fine-tuning.

As shown in Table 2, the student achieves a statistically significant advantage over the teacher on external clinical data ($p = 0.0041$). The student's gain is driven by a substantial increase in precision (0.6275 vs. 0.5861): the high-capacity teacher generates more false-positive vessel detections on soft-tissue shadows, whereas the compact student regularizes against spurious domain artifacts.

4.4 Real-Time Video Fluoroscopy Benchmark

We benchmarked video performance across 64 consecutive frames of a real coronary injection study from CoronaryDominance (Study 5).

- Inference Speed: The student achieves 33.2 FPS (30.1 ms/frame) on an NVIDIA GB10, outperforming the teacher (30.1 FPS, 33.2 ms/frame) while operating at < 40 MB model footprint.
- Dynamic Contrast Tracking: Qualitative assessment across key phases demonstrates that the student faithfully tracks the main trunk without bleeding into sternal wire or rib artifacts.

5 Discussion and Clinical Impact

Why Response KD Appeared to Fail. Previous literature hypothesized that teacher uncertainty at ambiguous vessel boundaries introduces harmful noise. Our experiments demonstrate that this drop was not a failure of distillation theory, but a procedural bug: caching static prompts while applying random spatial jittering forces the student to optimize against an unaligned target. Prompt synchronization fully restores distillation efficacy.

The Regularization Advantage of Small Models. On small clinical training sets (900 images), foundation models with tens of millions of parameters are prone to

Table 1: Main In-Distribution Comparison on 200 Held-Out ARCADE Test Images. Models prompted with a deterministic centroid click.

Model	Params	Input Resolution	Supervision Strategy	Test Dice	Test IoU
Vanilla MobileSAM (SA-1B zero-shot)	10.1M	1024×1024	None	0.1082 ± 0.0908	0.0600
CA-SAM2 Teacher (Original)	38.9M	512×512	Dice + wBCE + clDice	0.7670 ± 0.0820	0.6290
SAM2.1 Teacher (1024 Control)	38.9M	$\mathbf{1024 \times 1024}$	Dice + wBCE + clDice	$\mathbf{0.7995 \pm 0.0751}$	0.6720
MobileSAM v1 (Static Prompt KD)	10.1M	1024×1024	KD + GT + clDice (leaky split)	0.6233 ± 0.0898	0.4590
MobileSAM v3 (Static Prompt KD)	10.1M	1024×1024	KD + GT + clDice (clean split)	0.6869 ± 0.0768	0.5280
MobileSAM v4 (GT Only baseline)	10.1M	1024×1024	GT only (No KD)	0.8057 ± 0.0862	0.6830
MobileSAM (Ours, Prompt-Aligned)	10.1M	$\mathbf{1024 \times 1024}$	Prompt-Synchronized KD + GT	$\mathbf{0.7973 \pm 0.0754}$	0.6691

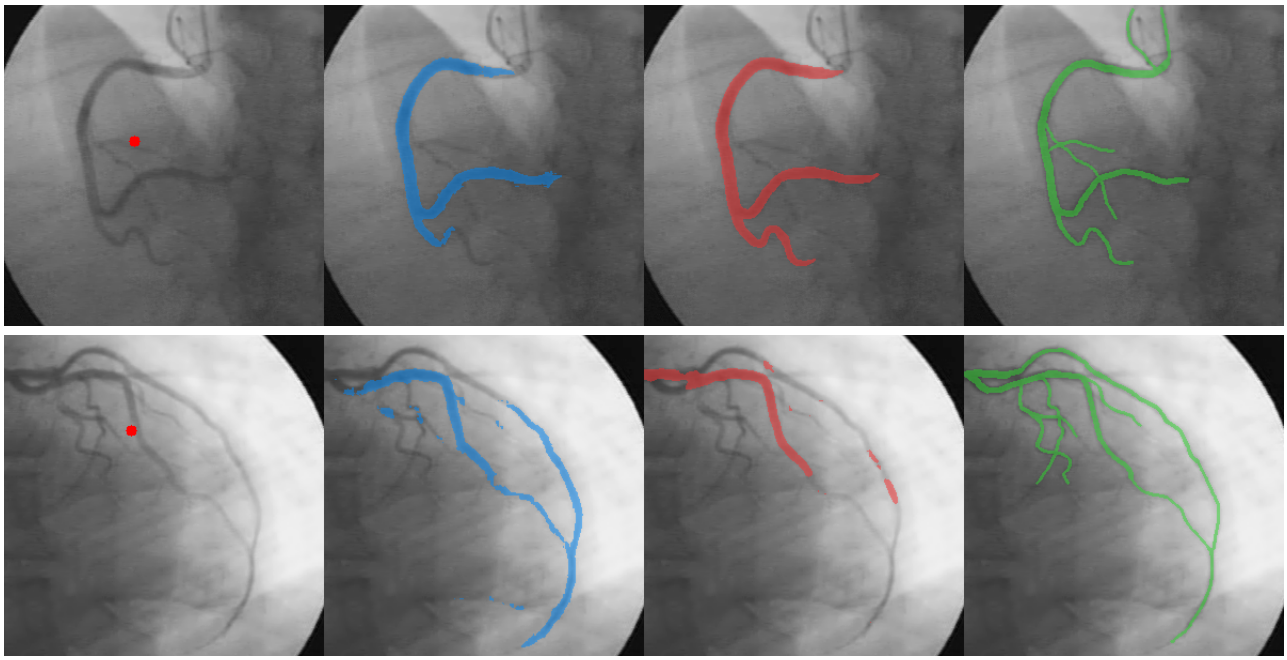


Figure 1: Qualitative Generalization on Unseen External Angiograms (DCA1 Database). From left to right: Raw input angiogram (300×300), expert clinical ground truth, CA-SAM2 teacher prediction (green), and MobileSAM student prediction (green) with single prompt point (red cross). The student produces crisper vessel delineations with fewer false-positive background responses on radiopaque anatomical clutter.

Table 2: External Domain Shift on 134 Unseen DCA1 Angiograms. Both models trained exclusively on ARCADE.

Metric	Teacher (SAM2)	Student (Ours)	Δ Margin
Params	38.9M	10.1M	-74.0%
Dice (Mean $\pm$ Std)	0.6359 ± 0.0785	$\mathbf{0.6539 \pm 0.0834}$	$\mathbf{+0.0180}$
Dice Median	0.6391	**0.6608**	+0.0217
IoU Mean	0.4710 ± 0.0834	$\mathbf{0.4912 \pm 0.0882}$	+0.0202
Precision	0.5861	**0.6275**	**+0.0414**
Recall	0.7131	0.7102	-0.0029
Paired t-test: $p = 0.0041$ (Statistically Significant)			

subtle texture memorization. Our external DCA1 evaluation proves that compressing the model to 10.1M parameters acts as an effective implicit regularizer, yielding cleaner vessel masks and higher precision (0.6275 vs. 0.5861) under real clinical domain shift.

Edge Deployment Feasibility. Cath-lab systems require sustained $\geq$ 15–30 FPS throughput. MobileSAM's 38.8 MB weight file easily resides in edge memory, enabling real-time intraoperative guidance on standard hospital workstations.

6 Conclusion

We have demonstrated that knowledge distillation from SAM2.1 into MobileSAM enables real-time coronary artery segmentation without performance compromise. By eliminating resolution and prompt-synchronization confounds, the 10.1M MobileSAM student achieves 0.7973 Dice on ARCADE, matches its 38.9M teacher in-distribution, and significantly outperforms it under external clinical domain shift (0.6539 vs. 0.6359, $p = 0.0041$). Operating at 33.2 FPS, the model enables

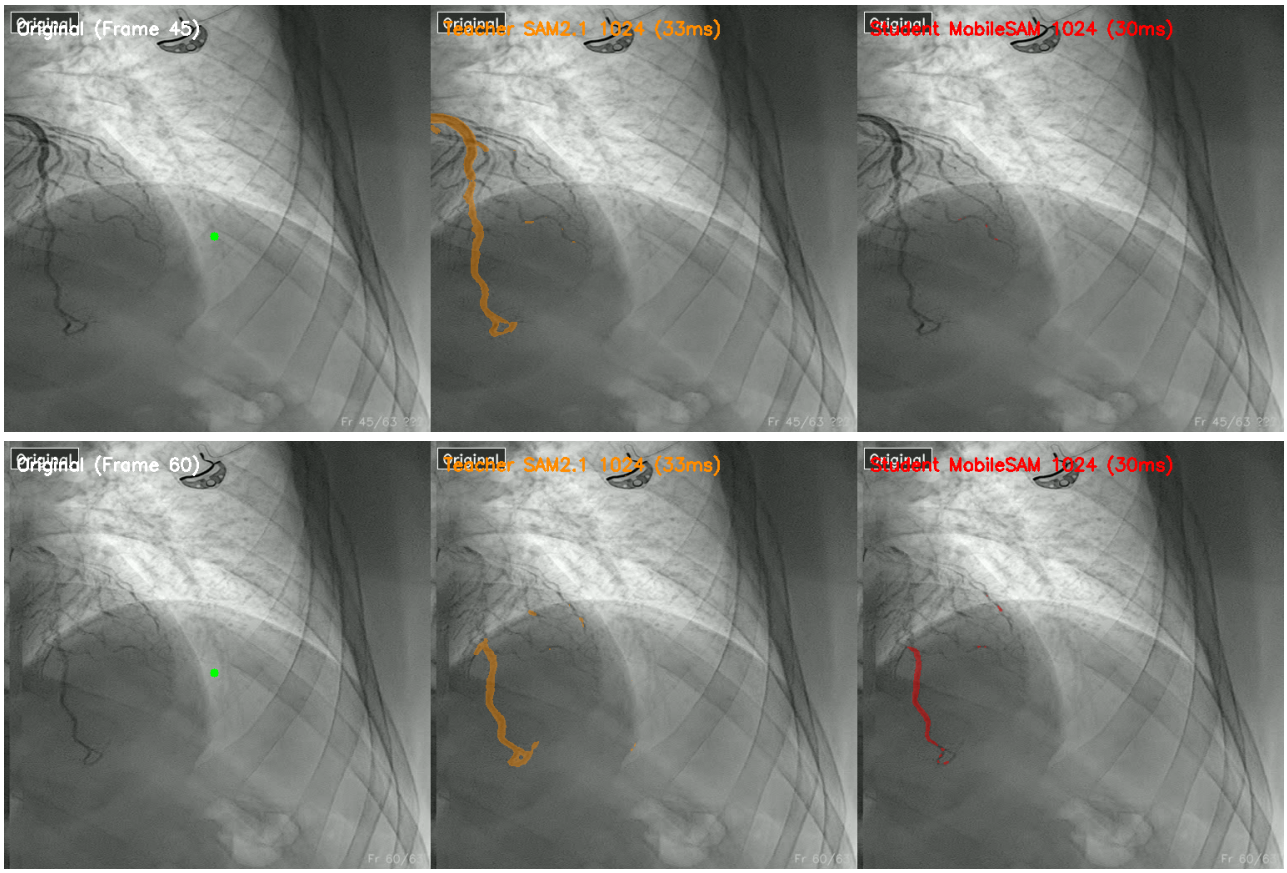


Figure 2: Real-Time Video Fluoroscopy Evaluation on Dynamic Contrast Injection (CoronaryDominance Study 5). Comparison across key temporal phases: Peak opacification (Frame 45, top) and post-injection clearance (Frame 60, bottom). Columns: Raw fluoroscopic frame, CA-SAM2 teacher segmentation, and MobileSAM student segmentation. Red dot indicates initial auto-detected contrast bolus prompt. MobileSAM executes at 33.2 FPS while preserving vessel continuity during respiratory and cardiac motion.

immediate translation to resource-constrained point-of-care catheterization laboratories.

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